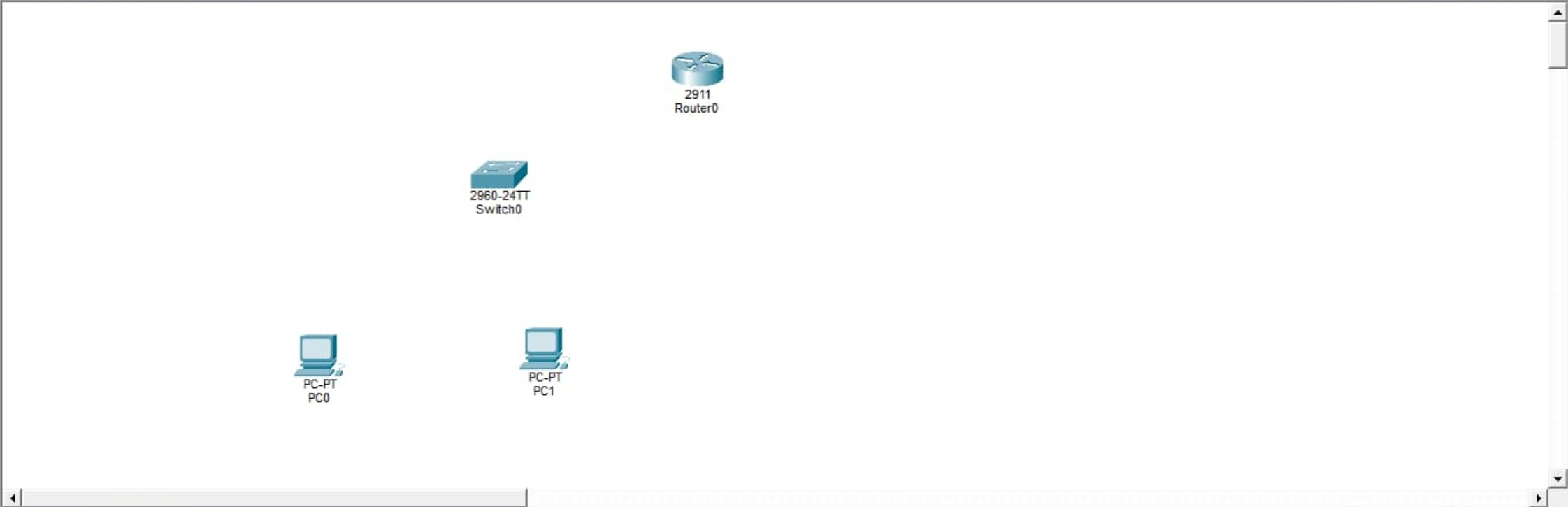




PC Laptop Server Wireless Server Network Controller Cyber Observer Data Historian Printer IP Phone VoIP Device Phone

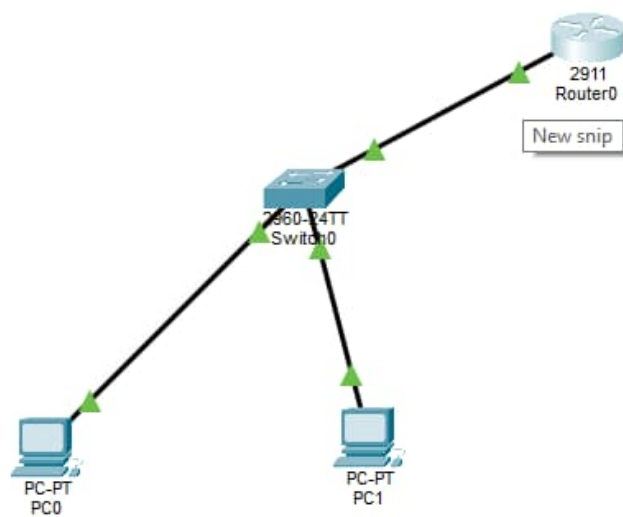
PC-PT

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
------	-------------	--------	-------------	------	-------	-----------	----------	-----	------	--------



Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
------	-------------	--------	-------------	------	-------	-----------	----------	-----	------	--------

Copper Straight-Through

IOS Command Line Interface

Switch	Ports	Model	SW Version	SW Image
-----	-----	-----	-----	-----
*	1 26	WS-C2960-24TT-L	15.0(2)SE4	C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: <http://www.cisco.com/techsupport>
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-S-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-S-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#vlan 10

Switch(config-vlan)#name ADMIN

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#name USER

Switch(config-vlan)#exit

Switch(config)#

IOS Command Line Interface

```
Switch>
Switch>
Switch>
Switch>
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name STAFF
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#do show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	STAFF	active	Fa0/1
20	ADMIN	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch(config)#
```

Copy

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IP Configuration

X

Interface

FastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.10

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::260:70FF:FE7E:D682

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

IP Configuration

X

Interface

FastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.11

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::201:C9FF:FE91:7608

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

IOS Command Line Interface

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface gigabitEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit

Router(config)#interface gigabitEthernet0/0

Router(config-if)#ip address 192.168.10.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#exit

Router(config)#interface gigabitEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#exit

Router(config)#exit

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.10.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router#

Copy

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Capturing from eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
4	101.885949950	192.168.56.103	192.168.56.1	ICMP	98	Echo (ping) request id=0x0002, seq=1/256, ttl=64 (reply in 5)
5	101.886464356	192.168.56.1	192.168.56.103	ICMP	98	Echo (ping) reply id=0x0002, seq=1/256, ttl=128 (request in 4)
6	102.897608536	192.168.56.103	192.168.56.1	ICMP	98	Echo (ping) request id=0x0002, seq=2/512, ttl=64 (reply in 7)
7	102.898393106	192.168.56.1	192.168.56.103	ICMP	98	Echo (ping) reply id=0x0002, seq=2/512, ttl=128 (request in 6)
8	103.899098738	192.168.56.103	192.168.56.1	ICMP	98	Echo (ping) request id=0x0002, seq=3/768, ttl=64 (reply in 9)
9	103.899722210	192.168.56.1	192.168.56.103	ICMP	98	Echo (ping) reply id=0x0002, seq=3/768, ttl=128 (request in 8)
10	104.912962090	192.168.56.103	192.168.56.1	ICMP	98	Echo (ping) request id=0x0002, seq=4/1024, ttl=64 (reply in 11)
11	104.913542609	192.168.56.1	192.168.56.103	ICMP	98	Echo (ping) reply id=0x0002, seq=4/1024, ttl=128 (request in 10)
12	105.937591862	192.168.56.103	192.168.56.1	ICMP	98	Echo (ping) request id=0x0002, seq=5/1280, ttl=64 (reply in 13)
13	105.938306991	192.168.56.1	192.168.56.103	ICMP	98	Echo (ping) reply id=0x0002, seq=5/1280, ttl=128 (request in 12)
16	106.938945947	192.168.56.103	192.168.56.1	ICMP	98	Echo (ping) request id=0x0002, seq=6/1536, ttl=64 (reply in 17)
17	106.939684737	192.168.56.1	192.168.56.103	ICMP	98	Echo (ping) reply id=0x0002, seq=6/1536, ttl=128 (request in 16)
20	107.940250875	192.168.56.103	192.168.56.1	ICMP	98	Echo (ping) request id=0x0002, seq=7/1792, ttl=64 (reply in 21)

Destination address

Frame 4: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bit) on interface 0

Ethernet II, Src: PCSSystemtec_23:b7:ee (08:00:27:23:b7:ee), Dst: 0a:00:00:00:00:00

Internet Protocol Version 4, Src: 192.168.56.103, Dst: 192.168.56.1

Internet Control Message Protocol

000 0a 00 27 00 00 06 08 00 27 23 b7 ee 08 00 45 00 ..'.....'#....E..

010 00 54 a9 f8 40 00 40 01 9e f7 c0 a8 38 67 c0 a8 ..T..@..@.....8g..

020 38 01 08 00 a9 eb 00 02 00 01 39 a2 71 69 00 00 8.....9qi..

030 00 00 de 32 06 00 00 00 00 00 10 11 12 13 14 15 ...2.....

040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25! "\$%&

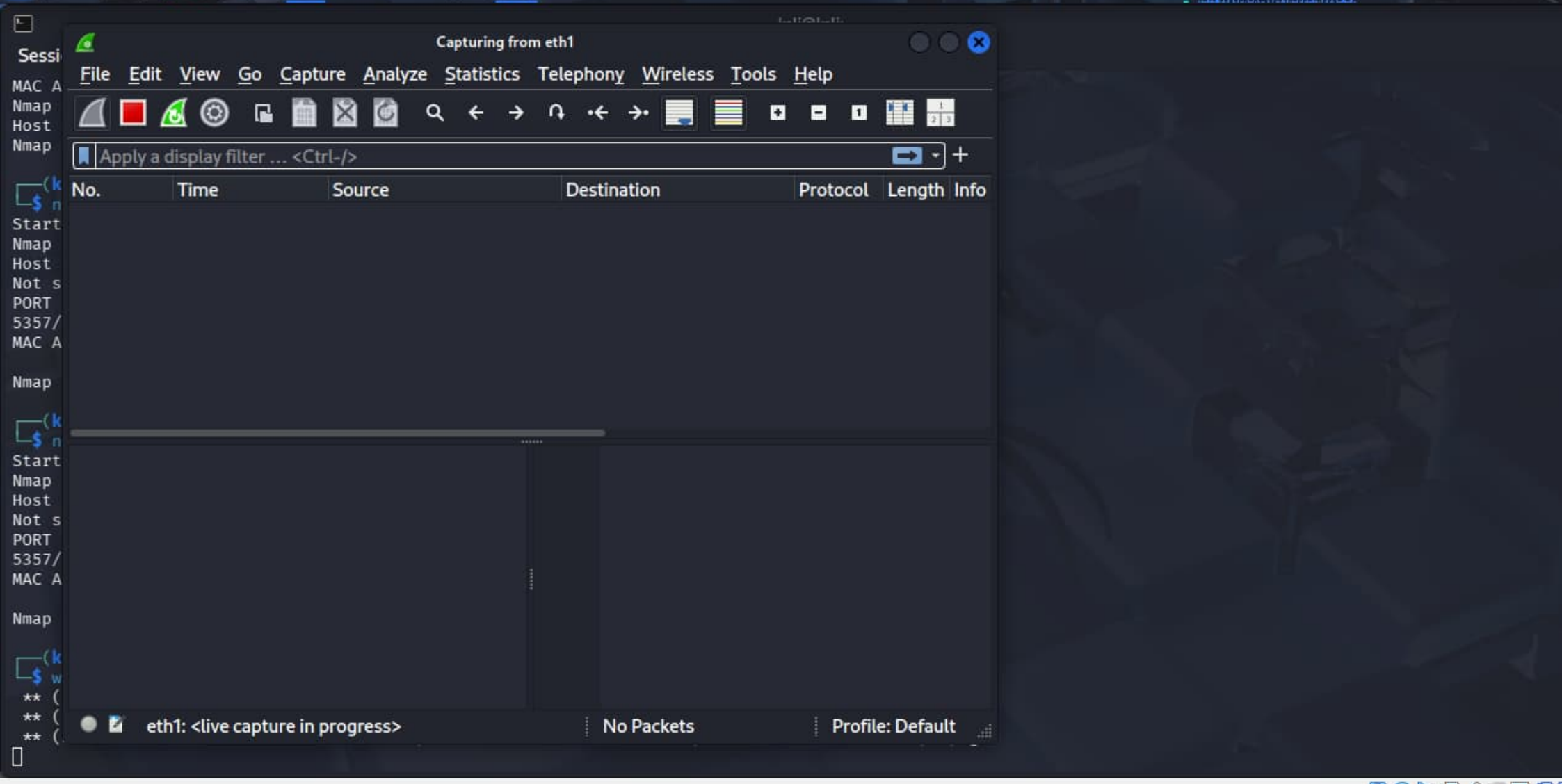
050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 &'()*+,-./012345

060 36 3767

Internet Control Message Protocol: Protocol

Packets: 93 · Displayed: 62 (66.7%)

Profile: Default



IOS Command Line Interface

```
SecureRouter#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SecureRouter(config)#username admin secret Admin@123
SecureRouter(config)#exit
SecureRouter#
%SYS-5-CONFIG_I: Configured from console by console

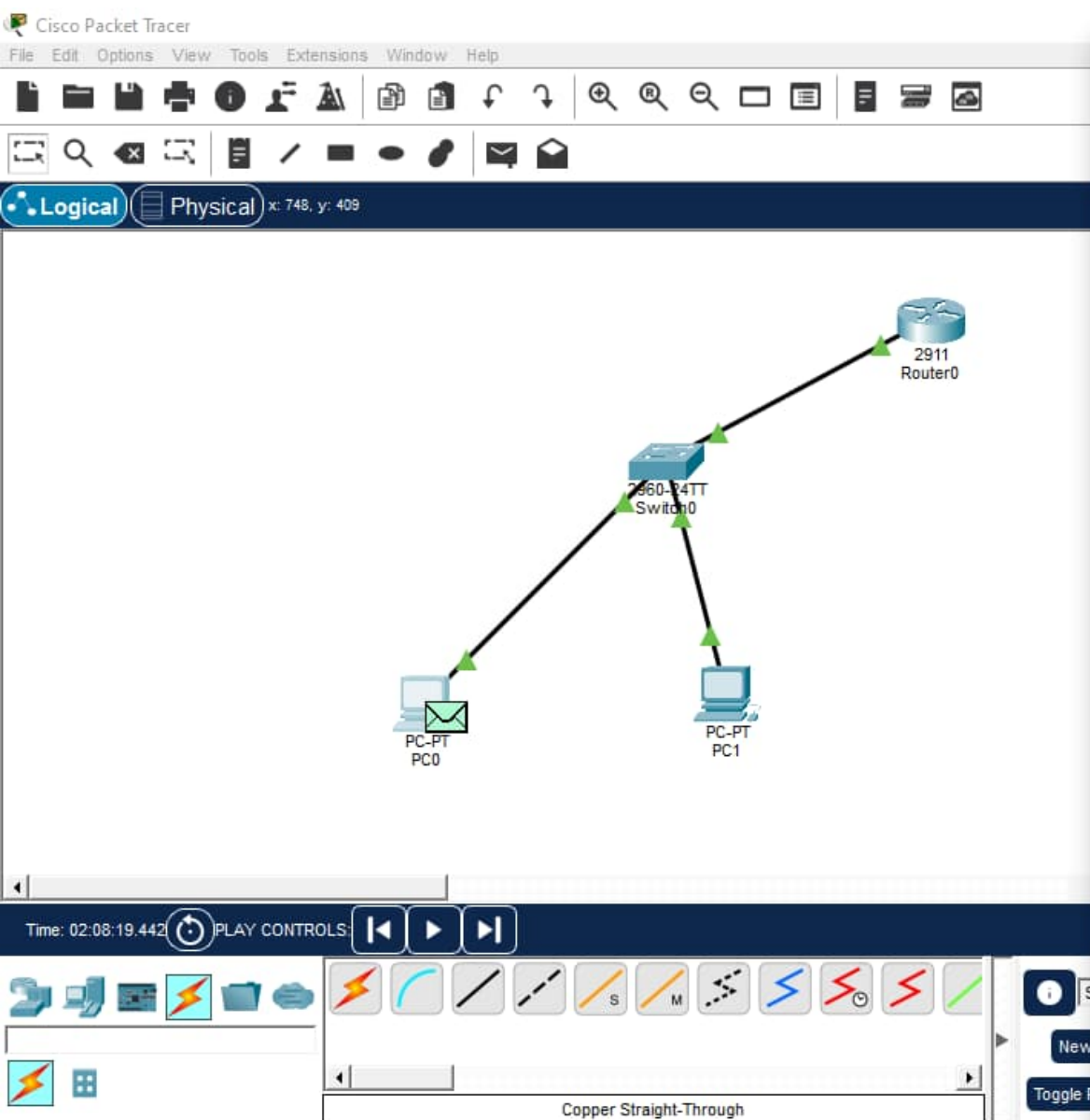
SecureRouter#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SecureRouter(config)#ip ssh version 2
SecureRouter(config)#exit
SecureRouter#
%SYS-5-CONFIG_I: Configured from console by console

SecureRouter#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SecureRouter(config)#line vty 0 4
SecureRouter(config-line)#login local
SecureRouter(config-line)#transport input ssh
SecureRouter(config-line)#exit
SecureRouter(config)#show ip ssh
^
% Invalid input detected at '^' marker.

SecureRouter(config)#do show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
SecureRouter(config)#do show running-config | section line vty
line vty 0 4
  login local
  transport input ssh
SecureRouter(config)#do show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
SecureRouter(config)#do show running-config | section line vty
line vty 0 4
  login local
  transport input ssh
SecureRouter(config)#
```

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PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=99ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 99ms, Average = 24ms

C:\>ssh -l admin 192.168.10.1

% Connection timed out; remote host not responding
C:\>ssh -l admin 192.168.10.1

% Connection timed out; remote host not responding
C:\>ssh -l admin 192.168.10.1

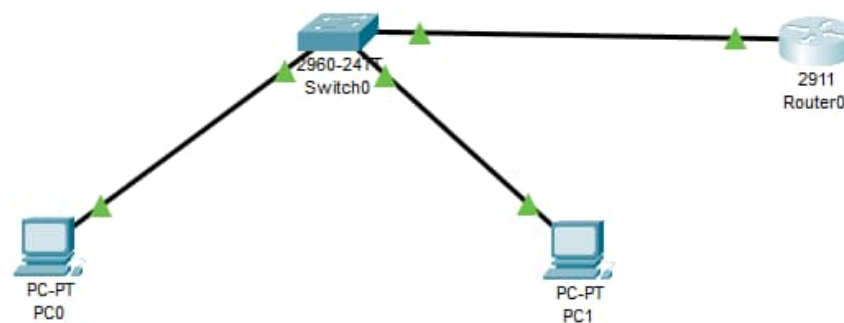
% Connection timed out; remote host not responding
C:\>
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
```



Logical Physical x: 886, y: 171

Root 03:52:00



Time: 00:07:42

Realtime Simulation



Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
------	-------------	--------	-------------	------	-------	-----------	----------	-----	------



IOS Command Line Interface

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no policy-map QOS-POLICY
Router(config)#no class-map IMPORTANT
Router(config)#no access-list 101
Router(config)#access-list 101 permit icmp any any
Router(config)#class-map match-all IMPORTANT
Router(config-cmap)#match access-group 101
Router(config-cmap)#policy-map QOS-POLICY
Router(config-pmap)#class IMPORTANT
Router(config-pmap-c)#priority 1000
Router(config-pmap-c)#exit
Router(config-pmap)#interface gigabitEthernet0/0
Router(config-if)#service-policy output QOS-POLICY
Router(config-if)#exit
Router(config)#do show policy-map interface gigabitEthernet0/0
GigabitEthernet0/0

Service-policy output: QOS-POLICY

Class-map: IMPORTANT (match-all)
  0 packets, 0 bytes
  5 minute offered rate 0 bps, drop rate 0 bps
  Match: access-group 101
  Queueing
    Strict Priority
    Output Queue: Conversation 264
    Bandwidth 1000 (kbps) Burst 25000 (Bytes)
    (pkts matched/bytes matched) 0/0
    (total drops/bytes drops) 0/0

Class-map: class-default (match-any)
  1 packets, 325 bytes
  5 minute offered rate 9 bps, drop rate 0 bps
  Match: any

Router(config)#
```

Copy

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Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Ping statistics for 192.168.20.2:

Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),



PC0

Physical Config Desktop Programming Attributes

Command Prompt

X

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=99ms TTL=255

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 99ms, Average = 24ms

C:\>|

☐ Top

```
(kali@kali)-[~]
```

```
$ ping 192.168.56.1
```

```
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data.
```

```
64 bytes from 192.168.56.1: icmp_seq=173 ttl=128 time=0.349 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=174 ttl=128 time=1.27 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=175 ttl=128 time=0.357 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=176 ttl=128 time=0.363 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=177 ttl=128 time=0.323 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=178 ttl=128 time=0.342 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=179 ttl=128 time=0.824 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=180 ttl=128 time=0.348 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=181 ttl=128 time=0.394 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=182 ttl=128 time=0.368 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=183 ttl=128 time=0.290 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=184 ttl=128 time=0.401 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=185 ttl=128 time=0.395 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=186 ttl=128 time=0.699 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=187 ttl=128 time=0.803 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=188 ttl=128 time=0.769 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=189 ttl=128 time=0.927 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=190 ttl=128 time=0.468 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=191 ttl=128 time=0.802 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=192 ttl=128 time=0.731 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=193 ttl=128 time=0.784 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=194 ttl=128 time=0.787 ms
```

```
64 bytes from 192.168.56.1: icmp_seq=195 ttl=128 time=0.681 ms
```

```
^C
```

```
— 192.168.56.1 ping statistics —
```

```
195 packets transmitted, 23 received, 88.2051% packet loss, time 198327ms
```

```
rtt min/avg/max/mdev = 0.290/0.586/1.273/0.254 ms
```

```
(kali@kali)-[~]
```

```
$
```

```
Connection-specific DNS Suffix  . :  
IPv6 Address. . . . . : 2401:4900:b9af:dc7b:1d24:9a89:8938:c497  
Temporary IPv6 Address. . . . . : 2401:4900:b9af:dc7b:7184:c8c3:c082:565  
Link-local IPv6 Address . . . . . : fe80::9ae2:82bc:e2b:a5bf%19  
IPv4 Address. . . . . : 10.19.0.76  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::44dd:b2ff:fe1f:2005%19  
                            10.19.0.172
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix  . :
```

Tunnel adapter Teredo Tunneling Pseudo-Interface:

```
Connection-specific DNS Suffix  . :  
IPv6 Address. . . . . : 2001:0:14c9:d700:4b4:adf6:e4c3:ca08  
Link-local IPv6 Address . . . . . : fe80::4b4:adf6:e4c3:ca08%14  
Default Gateway . . . . . :
```

```
C:\Windows\system32>netsh advfirewall firewall add rule name="Allow_Ping" dir=in action=allow protocol=ICMPv4  
Ok.
```

```
C:\Windows\system32>
```

```
C:\Windows\system32>
```



```
(kali@kali)-[~]
```

```
$ nmap -sn 192.168.56.0/24
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-21 22:39 -0500
```

```
Nmap scan report for 192.168.56.1
```

```
Host is up (0.00036s latency).
```

```
MAC Address: 0A:00:27:00:00:06 (Unknown)
```

```
Nmap scan report for 192.168.56.100
```

```
Host is up (0.00044s latency).
```

```
MAC Address: 08:00:27:80:84:E7 (Oracle VirtualBox virtual NIC)
```

```
Nmap scan report for 192.168.56.103
```

```
Host is up.
```

```
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.66 seconds
```

```
(kali@kali)-[~]
```

```
$ nmap -sS 192.168.56.1
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-21 22:42 -0500
```

```
Nmap scan report for 192.168.56.1
```

```
Host is up (0.00031s latency).
```

```
Not shown: 999 filtered tcp ports (no-response)
```

```
PORT      STATE SERVICE
```

```
5357/tcp  open  wsapi
```

```
MAC Address: 0A:00:27:00:00:06 (Unknown)
```

```
Nmap done: 1 IP address (1 host up) scanned in 15.99 seconds
```

```
(kali@kali)-[~]
```

```
$
```

Home

File System

Trash

VBox_GAs_...

kali@kali: ~

Session Actions Edit View Help

(kali@kali)-[~]

\$ ip a

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    aut qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86057sec preferred_lft 86057sec
    inet6 fe80::7302:f914:3e54:a5f6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:23:b7:ee brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixro
    ute eth1
        valid_lft 557sec preferred_lft 557sec
    inet6 fe80::4557:95d9:4e23:a96e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

(kali@kali)-[~]

\$

IOS Command Line Interface

```
SecureRouter(config)#ip ssh version 2
SecureRouter(config)#exit
SecureRouter#
%SYS-5-CONFIG_I: Configured from console by console

SecureRouter#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SecureRouter(config)#line vty 0 4
SecureRouter(config-line)#login local
SecureRouter(config-line)#transport input ssh
SecureRouter(config-line)#exit
SecureRouter(config)#show ip ssh
^
% Invalid input detected at '^' marker.

SecureRouter(config)#do show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
SecureRouter(config)#do show running-config | section line vty
line vty 0 4
  login local
  transport input ssh
SecureRouter(config)#do show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
SecureRouter(config)#do show running-config | section line vty
line vty 0 4
  login local
  transport input ssh
SecureRouter(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

SecureRouter(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       192.168.10.1    YES manual up              up
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
SecureRouter(config)#
```

Copy

Paste



16°C Cloudy

ENC
IN

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.20.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```
1004 addnet default          active
1005 trnet-default             active
Router#show class-map
  Class Map match-any class-default (id 0)
    Match any
  Class Map match-any ICMP_TRAFFIC (id 1)
    Match protocol icmp
Router#show policy-map
  Policy Map QOS-POLICY
    Class ICMP_TRAFFIC
  Policy Map QOS_POLICY
    Class ICMP_TRAFFIC
      Strict Priority
      Bandwidth 10 (%)
Router#show policy-map interface gigabitEthernet0/0
GigabitEthernet0/0

Service-policy output: QOS_POLICY

Class-map: ICMP_TRAFFIC (match-any)
  0 packets, 0 bytes
  5 minute offered rate 0 bps, drop rate 0 bps
  Match: protocol icmp
    0 packets, 0 bytes
    5 minute rate 0 bps
  Queueing
    Strict Priority
    Output Queue: Conversation 264
    Bandwidth 10 (%)
    Bandwidth 100000 (kbps) Burst 2500000 (Bytes)
    (pkts matched/bytes matched) 0/0
    (total drops/bytes drops) 0/0

Class-map: class-default (match-any)
  4 packets, 1232 bytes
  5 minute offered rate 26 bps, drop rate 0 bps
  Match: any

Router#
```

Ctrl+F6 to exit CLI focus

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Administrator: Command Prompt

Microsoft Windows [Version 10.0.19045.6466]
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C:\Windows\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter VirtualBox Host-Only Network:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::2681:fec4:b714:9728%6
IPv4 Address. : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter WiFi:

Connection-specific DNS Suffix . :
IPv6 Address. : 2401:4900:b9af:dc7b:1d24:9a89:8938:c497
Temporary IPv6 Address. : 2401:4900:b9af:dc7b:7184:c8c3:c082:565
Link-local IPv6 Address : fe80::9ae2:82bc:e2b:a5bf%19
IPv4 Address. : 10.19.0.76
Subnet Mask : 255.255.255.0
Default Gateway : fe80::44dd:b2ff:fe1f:2005%19
10.19.0.172

Ethernet adapter Bluetooth Network Connection:

IOS Command Line Interface

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name USER
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	ADMIN	active	Fa0/1
20	USER	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

```
\Windows\system32>
```

```
\Windows\system32>netsh advfirewall firewall add rule name="Allow_HTTP" dir=in action=allow protocol=TCP localport=80
```

```
.
```

```
\Windows\system32>
```

```
\Windows\system32>
```

IOS Command Line Interface

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.10.0.0.0.0.255 any access-list 100 deny
ip any any
      ^
% Invalid input detected at '^' marker.

Router(config)#access-list 100 permit ip 192.168.10.0.0.0.0.255 any
      ^
% Invalid input detected at '^' marker.

Router(config)#no access-list
% Incomplete command.
Router(config)#no access-list 100
Router(config)#access-list 100 permit ip 192.168.10.0.0.0.0.255 any
      ^
% Invalid input detected at '^' marker.

Router(config)#access-list 100 permit ip 192.168.10.0.0.0.0.255 any
      ^
% Invalid input detected at '^' marker.

Router(config)#
Router(config)#
Router(config)#no access-list 100
Router(config)#
Router(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255 any
Router(config)#access-list 100 deny ip any any
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#exit
Router(config)#do show access-lists
Extended IP access list 100
  10 permit ip 192.168.10.0 0.0.0.255 any
  20 deny ip any any

Router(config)#
```

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